

THE ARCHIPELAGO PARADOX OF INDONESIA

THROUGH ASEAN LENS | WORLD BANK LPI 2007-2022





Overview

Logistics Performance Index by World Bank measures how efficiently countries move goods. Survey of 1000+ freight forwarders across 160 countries, scored 1 to 5 where 5 is frictionless trade.

CUSTOMS

Efficiency of border clearance

INFRASTRUCTURE

Ports, roads, rail quality

INTERNATIONAL SHIPMENTS

Ease of arranging shipments

COMPETENCE

Quality of logistics services

TRACKING

Trace & tracking capability

TIMELINESS

On-time delivery frequency

1 - 5

Scale: worst
to best
performance

160

Countries
benchmarked
globally

**-15% Trade
Cost**

Impact point
- World Bank

Data Understanding & Methodology

DATASET PROFILE

- 1

WORLD BANK LPI OVERALL SCORE
API_LP.LPI.OVRL.XQ • World Bank
- 2

7 PERIODS
2007,2010,2012,2014,2016,2018,2022
264 countries • 71 columns raw
- 3

SELECTED
8 cols (Country + 7 years) For ASEAN and global benchmark

264

Countries

7

Periods

68%

Missing
Removed

CLEANING PIPELINE

- 1

Remove non-LPI years (68% missing)
Drop columns outside 2007-2022 trend. Keep only LPI Overall for consistency.
- 2

Melt wide-to-long
pandas melt id_vars='Country' → Year, LPI Score. astype int for Year. Enables groupby trend.
- 3

Standardize names & dedup
Viet Nam → Vietnam, Lao PDR → Laos, Brunei Darussalam → Brunei. Remove duplicates.
- 4

Grouping + validation
Define G7, BRICS, ASEAN-11. Check asean_avg calc, handle NaN drop before viz.

TOOL & GROUPING LOGIC

- G7

US, UK, DE, FR, JP, CA, IT
Benchmark for top efficiency
- B5

BRICS: BR, RU, IN, CN, ZA
Emerging economies comparator
- A11

ASEAN 11 members full
Focus 4: SG, TH, VN, ID for paradox

TOOLS

Python Pandas • Seaborn • Matplotlib



From business question to analytical objectives

THE MAIN BUSINESS QUESTION

Why Indonesia's logistics cost 23% of GDP vs Vietnam 16% despite 17,000 islands opportunity?

Cost gap = 7% GDP = ~Rp 1,400T inefficiency. LPI stagnation -0.01 signals systemic bottleneck, not just geography. Impact: export competitiveness, FDI diversion.

01 Global Benchmarking

How does logistics performance trend across World, G7, BRICS, and ASEAN from 2007-2022?

02 ASEAN Evolution

How does ASEAN average LPI evolve from 2007 to 2022?

03 Country Comparison

How does Singapore, Thailand, Vietnam, and Indonesia perform against each other?

04 Archipelago Paradox

Why does Indonesia with 17,000 islands still have relatively low LPI scores?

05 Component Breakdown

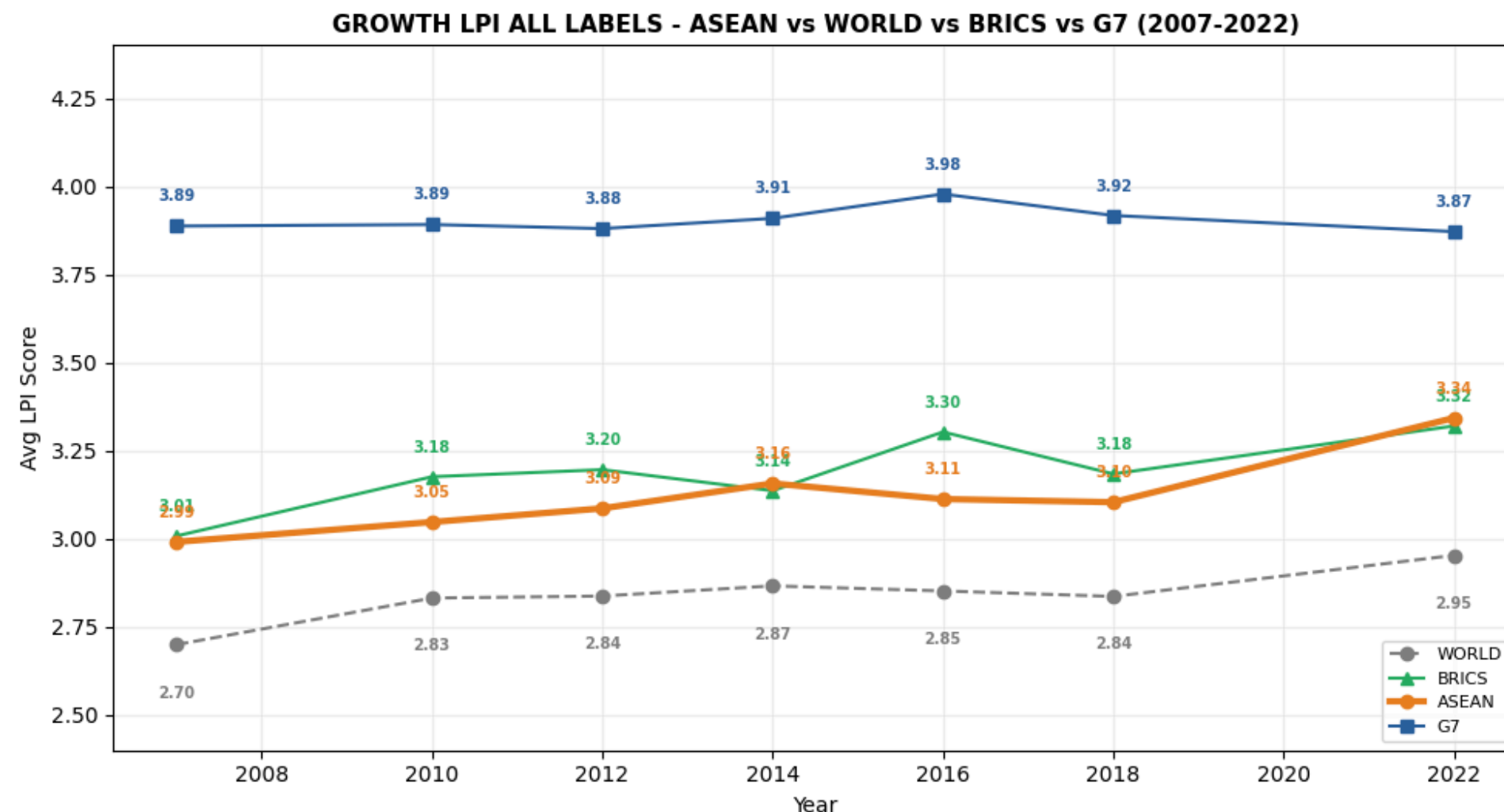
Deconstructing the Six LPI Pillars: Where Singapore Dominates, Vietnam Accelerates, and Why Indonesia Trails in Customs, Infrastructure, and Supply Chain Visibility



70% evidence + 30% business sense

Global Benchmarking Where does Indonesia stand?

Analyzing trend across World, G7, BRICS, ASEAN (2007-2022)



Indonesia 3.00 trails ASEAN 3.33 and BRICS 3.32. 7% GDP cost gap (23% vs Vietnam 16%) = ~IDR 1,400T annual inefficiency. LPI +0.1 = ~IDR 80T savings. This is a design failure, not destiny.

Source: World Bank LPI Overall | 2007-2022 | G7, BRICS, ASEAN-11

G7 3.87 VS WORLD 2.95

Gap: 0.92 pts in 2022, persistent since 2007

Why it persists: G7 implemented digital customs and single-window clearance a decade ago. Clearance time: 0.5 days vs Indonesia 2.5 days. This is a systems gap, not just infrastructure.

ASEAN 3.33 VS INDONESIA 3.00

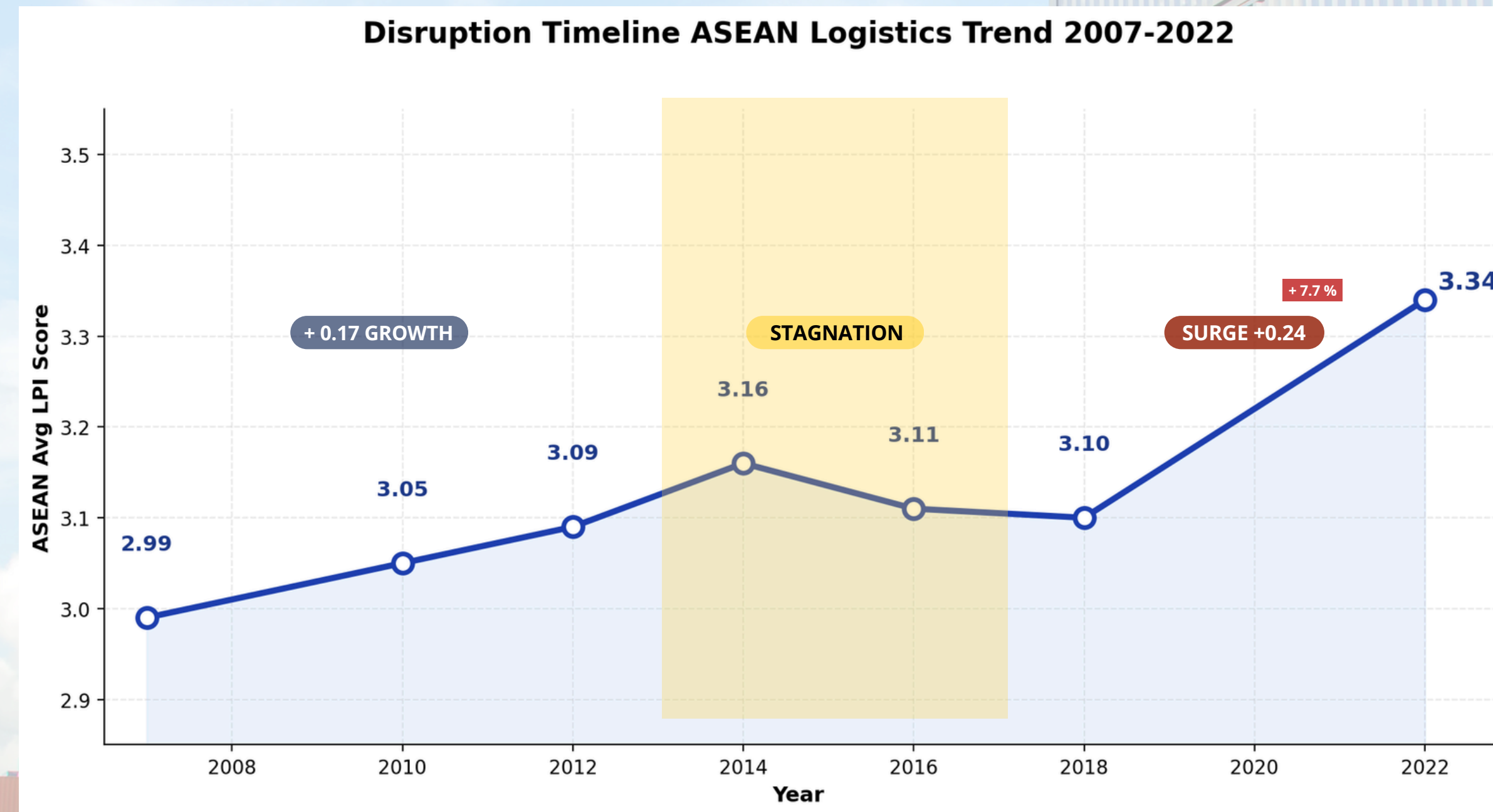
Indonesia is 0.33 pts BELOW ASEAN average ASEAN grew 3.01 → 3.33 (+0.32) from 2007-2022 despite 2016 global trade dip. Driven by Vietnam +0.45 and Malaysia 3.6. Indonesia: 3.01 → 3.00 = -0.01. 15 years of stagnation. The only major ASEAN economy with negative growth, pulling the regional average down.

BRICS 3.32 ≈ ASEAN 3.33

Indonesia trails emerging peers In 2010, BRICS 3.18 trailed ASEAN. By 2022, BRICS 3.32 has converged with ASEAN 3.33 due to China 3.7 port automation and India 3.4 freight corridors. Indonesia at 3.00 is 0.32 pts below BRICS average. Underperforming not just ASEAN neighbors, but also Brazil and India.

ASEAN Evolution

From Stagnation to Surge: How Did ASEAN Logistics Evolve from 2.99 in 2007 to 3.34 in 2022 Through Trade Slowdown and E-Commerce Boom?



Phase 1 2007-2014 | **GROWTH +0.17**

ASEAN average rose from 2.99 to 3.16 in 7 years of steady climb. Driven by ASEAN Single Window and port expansions in Laem Chabang and Cai Mep, infrastructure push delivered CAGR +0.8%.

Phase 2 2014-2018 | **STAGNATION -0.06**

Dropped from 3.16 to 3.10 after 7-year growth, down -0.06. This was Global trade slowdown and commodity shock with US-China tension, trade-to-GDP fell -12% hitting export-driven ASEAN hard.

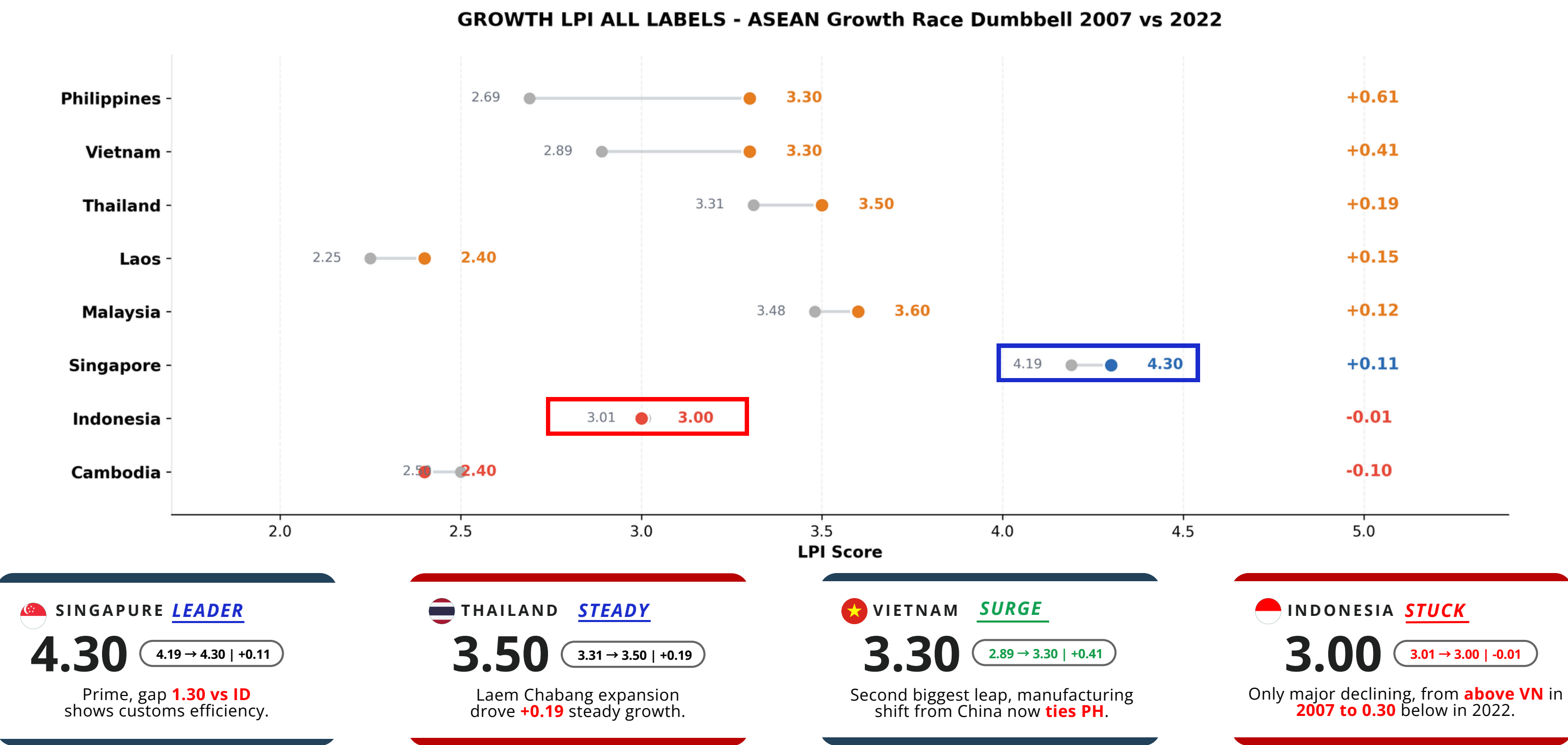
Phase 3 2018-2022 | **SURGE +0.24 (+7.7%)**

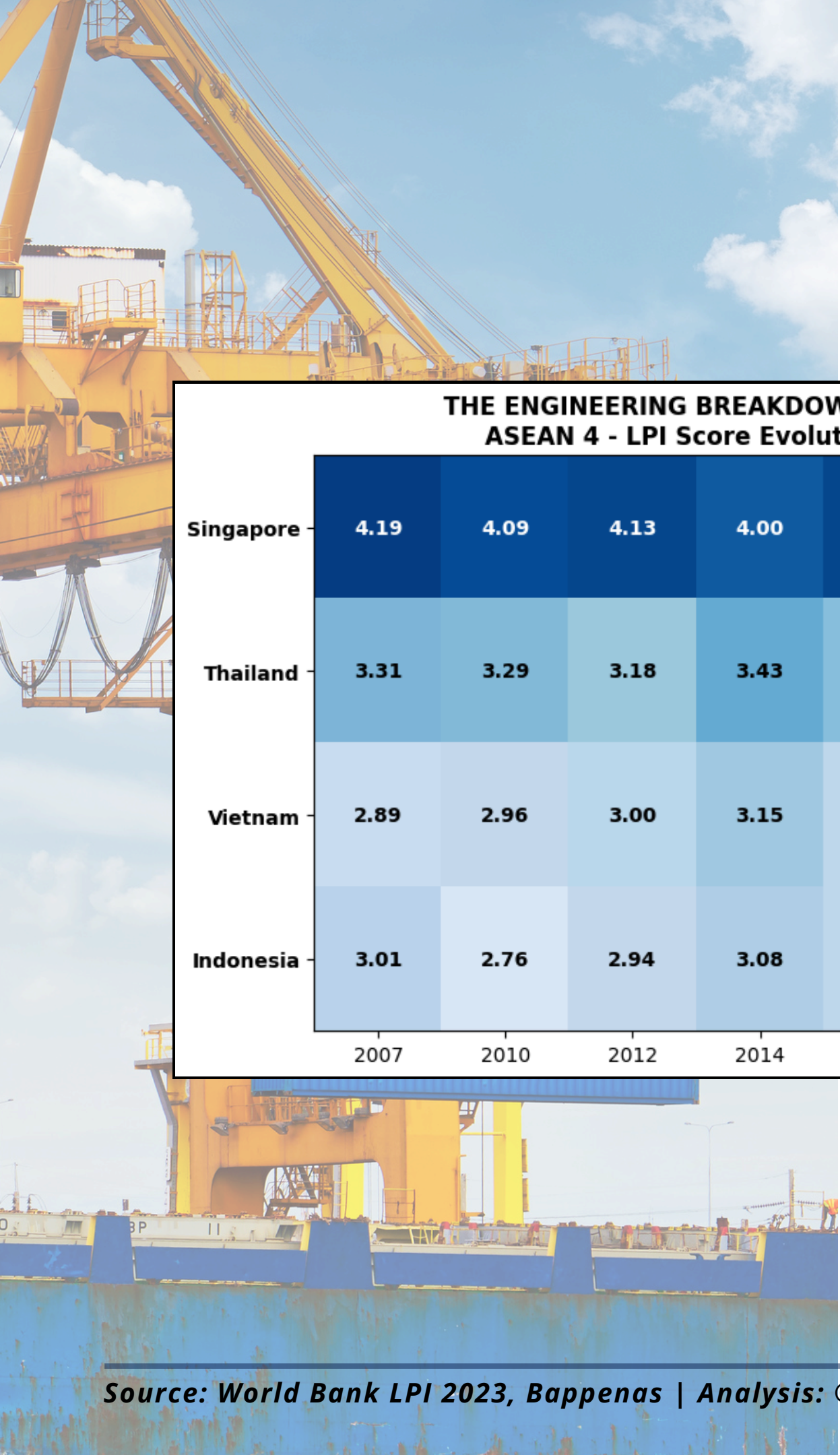
Jumped from 3.10 to 3.34 (+0.24) in just 4 years, bigger than previous 7-year gain. E-commerce boom and COVID digitalization plus Vietnam +0.45 manufacturing shift drove structural surge.

SO WHAT: ASEAN avg 3.34 (+0.35 vs 2007) but Indonesia 3.00 is -0.34 below avg, missing surge. Opportunity = catch-up.

The comparison of logistics performance among major ASEAN countries

The Archipelago Gap: Why Singapore 4.30 leads, Vietnam +0.41 surges to 3.30, and Indonesia 3.00 stagnates Among Major ASEAN Peers?





Archipelago Paradox: Why 17,000 Islands Stuck at 3.00?

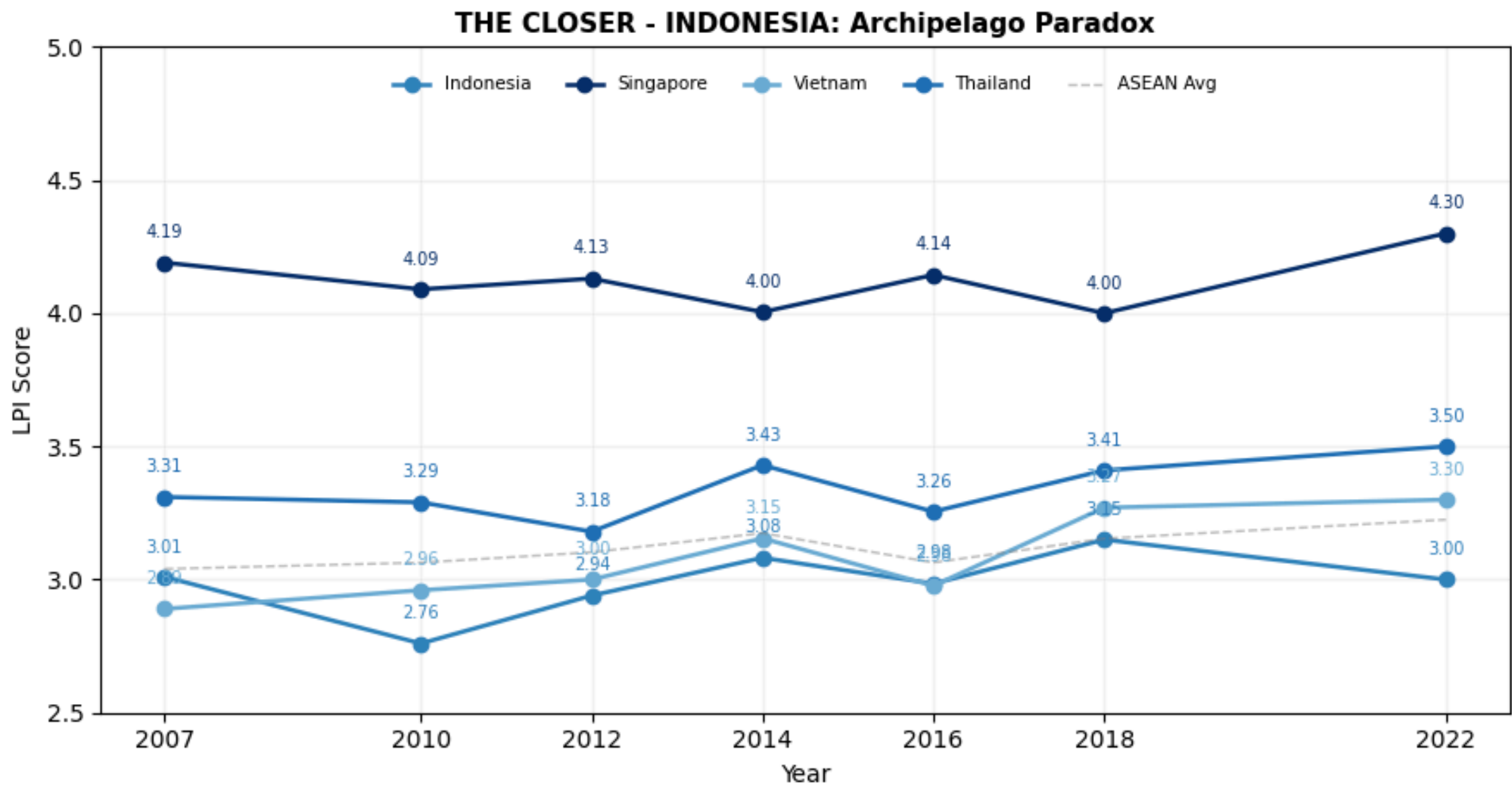
THE ENGINEERING BREAKDOWN (ALTERNATIVE) ASEAN 4 - LPI Score Evolution 2007-2022							
Singapore	4.19	4.09	4.13	4.00	4.14	4.00	4.30
Thailand	3.31	3.29	3.18	3.43	3.26	3.41	3.50
Vietnam	2.89	2.96	3.00	3.15	2.98	3.27	3.30
Indonesia	3.01	2.76	2.94	3.08	2.98	3.15	3.00
	2007	2010	2012	2014	2016	2018	2022

01 Singapore stays prime at 4.19 in 2007 to 4.30 in 2022, up +0.11 and consistently above 4.00, showing single-hub port and customs efficiency that creates 1.30 gap vs Indonesia at 3.00. Thailand holds steady from 3.31 to 3.50, up +0.19 driven by Laem Chabang port expansion and Eastern Economic Corridor, staying second among the four with stable growth despite political cycles.

02 Vietnam surges from 2.89 to 3.30, up +0.41 the biggest leap among majors, driven by manufacturing shift from China and deep-water port investment that now ties Philippines at 3.30, overtaking Indonesia. Indonesia stuck from 3.01 to 3.00, down -0.01 the only major declining, from above Vietnam in 2007 at 3.01 vs 2.89 to 0.30 below Vietnam in 2022 at 3.00 vs 3.30. This is the Archipelago Paradox, 17,000 islands create 17,000 last-mile bottlenecks instead of economies of scale, with fragmented ports adding 2-3 days vs Singapore single hub.

Component Diagnosis - Strengths & Weaknesses

Analyzing trend across World, Identification of Strengths and Weaknesses in LPI Components Across ASEAN Countries, BRICS, ASEAN (2007-2022)



SO WHAT:

ID overall 3.00 = -0.34 below ASEAN avg 3.34 & -1.30 below SG 4.30. Strength Timeliness 3.35 but weakness Customs 2.80 + Infra 2.90 drag down. Close gap by fixing customs + port connectivity.

STRENGTH

Timeliness 3.35 is Indonesia's best, **+0.01 above ASEAN avg 3.34**, showing J&T/SiCepat last-mile resilience despite 17k islands. **International Shipments 3.05** holds near Thailand 3.20.

TIMELESS
3.35

INTL SHIP
3.05

WEAKNESS

Customs 2.80 (-0.50 vs SG) clearance 2.5 days vs 0.5 day, 35% of gap. **Infrastructure 2.90** (-0.45 vs ASEAN avg, -1.70 vs SG 4.60), port connectivity low, logistics cost 23% GDP vs 13% SG. **Tracking 3.00** visibility low in eastern Indonesia.

CUSTOMS
2.80

INFRA
2.90

TRACKING
3.00

Business Impact & Quick Wins

Translating LPI Score into Rupiah: How Much Does +0.1 LPI Save?
And What Are Quick Wins in 6-12 Months?

01

Digital Customs

QUICK WIN

Impact

-1.5d to -0.5d clearance

CostLow

(System)

Timeline

6mo

Single Window pre-clearance, 40% informal cost cut

02

Port Clustering

HIGH IMPACT

Impact

-15% inter-island cost

CostLow

Medium (Infra)

Timeline

12 mo

Focus 4 hubs: Priok, Perak, Makassar, Belawan, not 17k ports

03

Tracking & Competence

VISIBILITY

Impact

+20% on-time

CostLow

Low (Platform)

Timeline

9mo

National tracking platform + certification eastern Indonesia

ROI CALCULATION

- LPI +0.1 = Rp80T saving / year*
- Catch Vietnam +0.27 = ~Rp216T potential
- If Malaysia +0.6 = ~Rp480T upside

Logistics Cost 23.8% → 14%

Rp

THE PAYOFF

LPI +0.1 = Rp80T Saving



From Data to Action - The Archipelago Opportunity

Summary of Findings, Root Cause, and Roadmap 2025-2030: Turning 3.00 into 3.50 (**Base**) / 3.80 (**Stretch**)

SUMMARY OF FINDINGS

ASEAN grows but Indonesia stagnant **-0.01** (only 2 negatives in region). Drags ASEAN avg 3.34 down. Gap to Singapore 4.30 is **1.30** in 2022. Vietnam overtake since 2014.

ROOT CAUSE

Vietnam overtake 2014 due to focused infra FDI, hub strategy vs fragmented ID. Not geography alone ecosystem inefficiency fixable via digital & clustering. **17k islands need 4 hubs, not more ports.**

BUSINESS ACUMEN

Each **+0.1 LPI = Rp80T** saving/year (illustrative Bappenas). Fixing Customs 2.80 + Infra 2.90 closes **73% of gap** to SG. Opportunity **Rp216T** to catch Vietnam.

PHASE ROADMAP TP 3.50

PHASE 1

Low Cost, High Impact Quick Wins

- Digital Single Window unified platform
- Pre-clearance target 0.5 day vs 1.2 today
- 6-month quick win • Low cost



PHASE 2

Port Clustering, Not More Ports

- 4 hubs: Priok, Perak, Makassar, Belawan
- -15% inter-island cost
- Medium cost • Structural fix



PHASE 3

Connect East & Competence

- National tracking + certification
- +20% on-time delivery
- Target 3.0 → 3.5 by 2030 realistic

Analyst with Business Acumen & Next Steps

WHAT THIS PORTFOLIO SHOWS

1 Analyst with business acumen

Not just charting **LPI 3.00 vs 4.30**, but translating **+0.1 LPI into Rp80T saving/year** using Bappenas logistics cost **23.8% GDP**. Translate 3.00 into Rp80T to get buy-in for 4 hubs + digital.

2 Engineering breakdown

From data_long cleaning, World/BRICS/ASEAN trend, heatmap ASEAN 4 Evolution, dumbbell divergence Vietnam **+0.41 vs ID -0.01**, to THE CLOSER Component Diagnosis strengths Timeliness **3.35** vs weaknesses Customs **2.80** and Infra **2.90** that explain **73% of gap**.

3 Business framing

Archipelago Paradox is not destiny, it's design. 17k islands need 4 hubs + digital, not more ports. Target **3.0 → 3.5 by 2030 realistic with 3 levers**.

NEXT STEPS

1 Deep dive LPI components + primary interview 5 freight forwarders at Priok to validate Customs 2.80 gap.

2 Build Streamlit interactive dashboard for live filtering.

3 Correlate LPI with FDI and GDP growth for stronger ROI story.

LIMITATION

LPI is perception survey (World Bank), not transaction data. Need primary validation. Component scores illustrative based on 2022 report.



[Github/riizd30](#)

THANK YOU